Policy Analysis of the Affordable Health Care Act (ACA) on Premiums per Person

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Abstract

We are looking at the impact of the Affordable Care Act (ACA) on health insurance premiums in the United States. Using data from the US Department of Labor, our project employs a difference-in-difference (DiD) approach to compare 40 states that adopted the ACA with 10 that did not. The analysis focuses on per capita premiums for both experience-rated and non-experience-rated contracts. The main goal of this project is to assess the effectiveness of the Act in improving healthcare affordability for average Americans.

Variables 1

VARIABLE	Units	Mean of Means	Mean of Standard Deviations	Overall Minimum	Overall Maximum
INS_PRSN_COVERED_EOY_CNT (Insured Persons Covered End of Year Count)	Number of people	1333.67	42041.70	0.00	9999999.00
WLFR_PREMIUM_RCVD_AMT (Welfare Premium Received Amount. It represents the total premiums received by the insurance company for the coverage provided during the policy year. This is only for experience-rated contracts.)	US Dollars	2394116.44	937505906.19	-130923183.00	8701261226322.00
WLFR_TOT_CHARGES_PAID_AMT (Welfare Total Charges Paid Amount. It represents the total amount of charges paid by the insurance company for claims during the policy year. This is for non-experience rated contracts.)	US Dollars	2900620.40	1250374247.52	-50876783.00	7708480105986.00

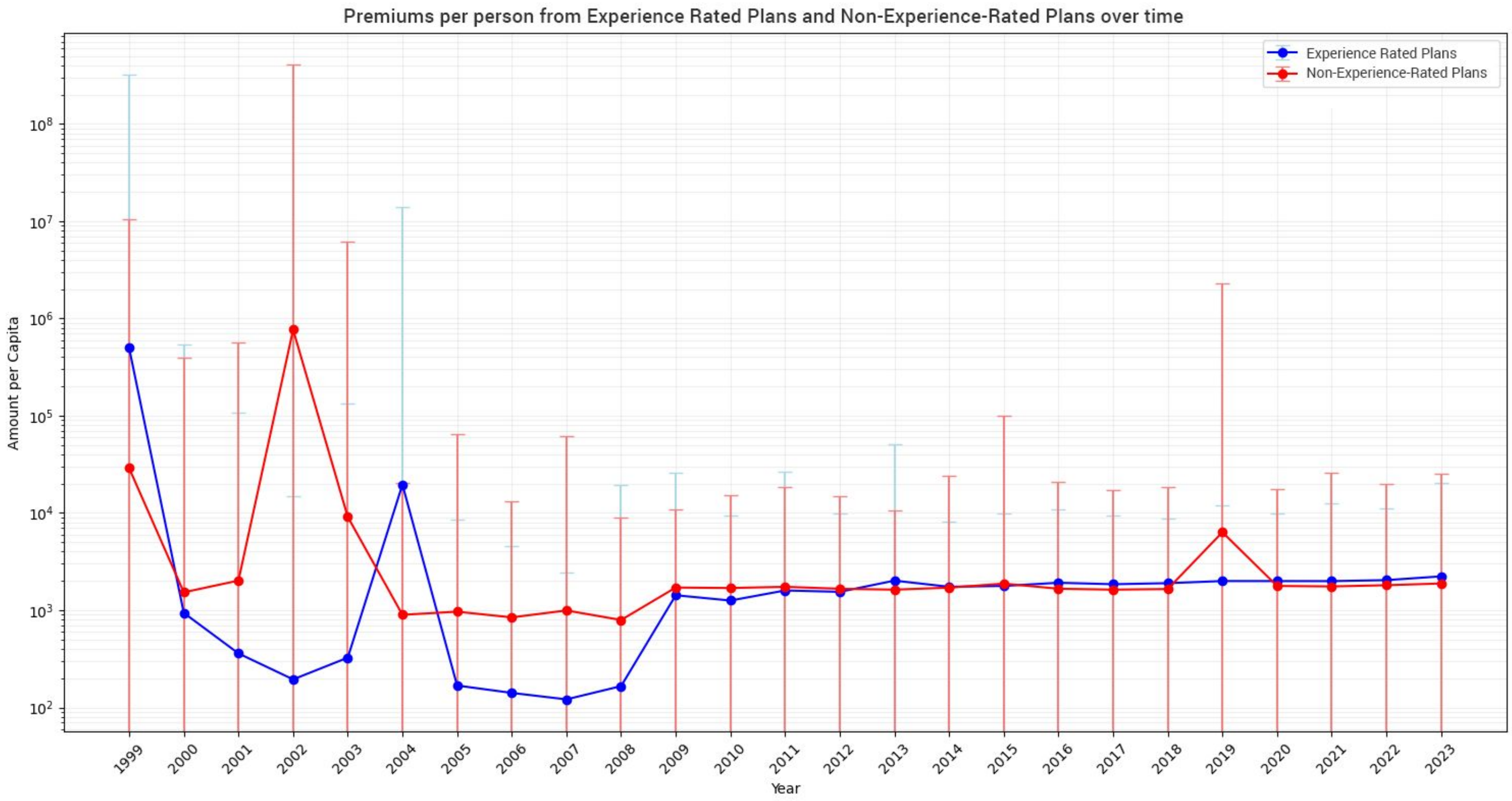
Variables 2

VARIABLE	Units	Mean of Means	Mean of Standard Deviations	Overall Minimum	Overall Maximum
SPONS_DFE_STATE (This variable represents the plan sponsor / Direct Filing Entity (DFE). It indicates the state where the plan sponsor or DFE is located.)	two-letter state IDs				
WLFR_PREMIUM_RCVD_AMT_PER_CAPITA (Dividing WLFR_PREMIUM_RCVD_AMT by INS_PRSN_COVERED_EOY_CNT)	Number of people	21864.86	13302903.79	-178856.81	204126900569.05
WLFR_TOT_CHARGES_PAID_AMT_PER_CAPITA (Dividing WLFR_TOT_CHARGES_PAID_AMT by INS_PRSN_COVERED_EOY_CNT)	US Dollars	33723.96	16920728.97	-4771105.50	211823456790.11
ID (plan Identification number)	Numeric Code				

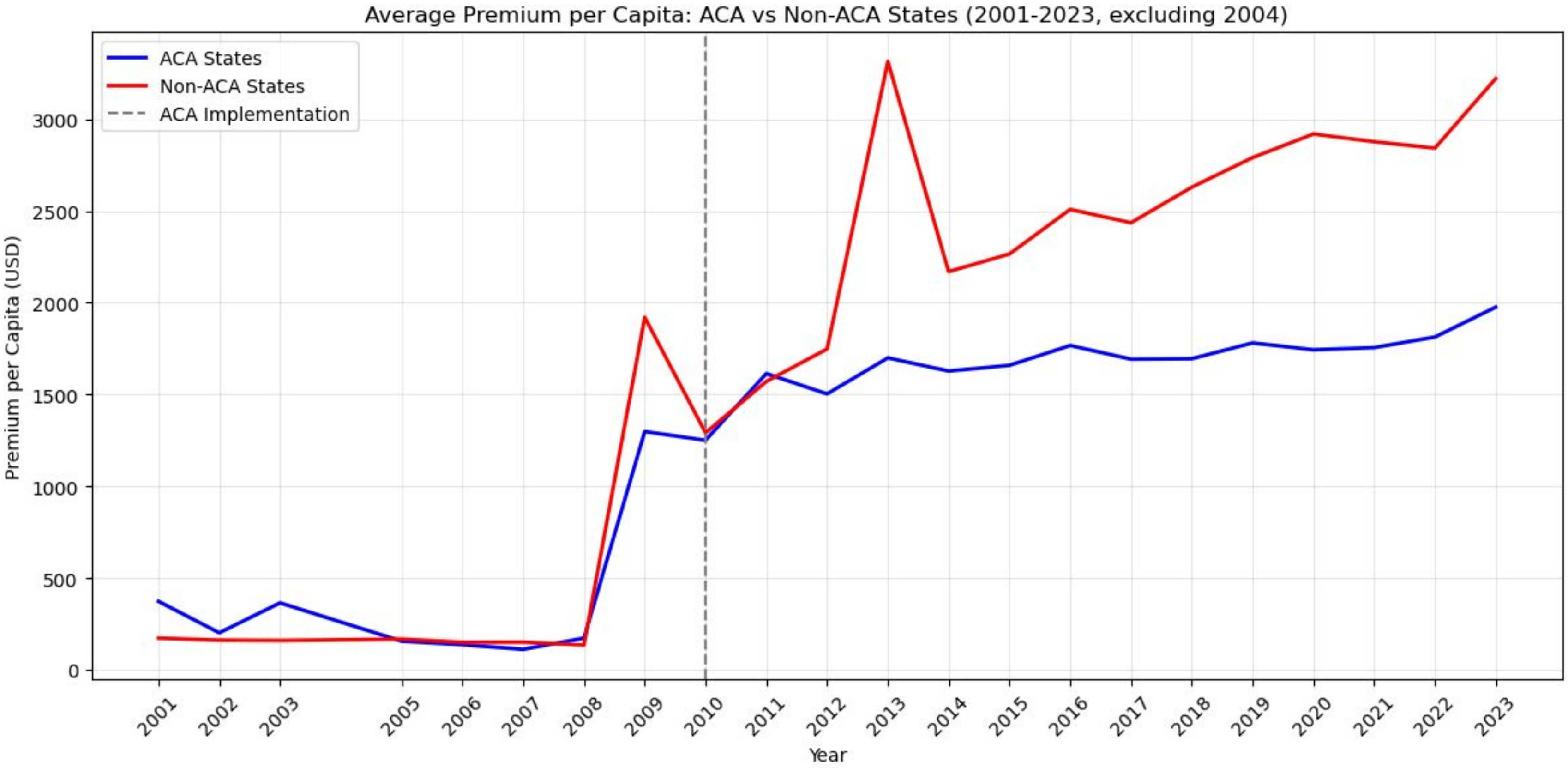
Control Variables

VARIABLE	Units	Mean	Standard Deviations	Overall Minimum	Overall Maximum
Inflation	Percentage	2.55%	1.63%	-0.36%	8.00%
Inflation Annual Change	Percentage	0.10%	1.76%	-4.19%	3.46%
Market Indicator (S&P 500 Yearly Growth)	Percentage	7.1780%	17.7607%	-38.49%	29.6%
Federal Reserve Interest Rates ()	Percentage	1.91%	1.99%	0.08%	6.24%

Variable over time



Parallel Trends



Methodology

The method we are going to use for our analysis is a difference-in-difference (DiD) regression to find the causal effect of the Affordable Care Act (ACA) on insurance premiums per person for both Experience Rated and Non-Experience Rated plans. We have decided on this approach for three reasons.

- The implementation of the ACA creates a natural experiment where 40 states implemented it and 10 did not.
- We had data from 1999-2023 and ACA was implemented in 2010 giving us long term post and pre-treatment data.
- The implementation is also clear with the ACA becoming implemented in 2010.

Regression Equation

$$Y_{ist} = \beta_0 + B_1 ACA_s + \beta_2 Post + \beta_3 (ACA_s * Post_t) + \gamma Controls_t + \varepsilon_{ist}$$

Y_{ist} - represents premiums per capita for either experience based plans or non experience-based plans. i is plan,state is s and time is t.

ACA_s - dummy for if the state adopted the ACA

$Post_t$ - is a dummy for post-2010 periods

$Controls$ - represents a vector of all of our controls (inflation, Market indicator (S&P 500 growth), Federal Interest Rates)

ε_{ist} - error term

As for Data cleaning for this regression to work we had to

- Omit the years 1999, 2000 and 2004 because of the large variance and lack of data for these early years.
- We also trimmed of 1% of outliers which we saw had to be dealt with from our minimums and maximums.
- **Log Transform Y (MAYBE)**

Results of Regression (Without Controls)

Experience Rated Plans	Coefficient	Std Error	t-stat	p-value	CI Lower	CI Upper
Constant	6.871	0.019	355.624	~0	6.833	6.908
ACA_State	-0.209	0.023	-9.271	~0	-0.254	-0.165
Post_ACA	0.868	0.023	37.119	~0	0.822	0.914
ACA_x_Post	-0.093	0.029	-3.178	0.0015	-0.150	-0.035

Non-Experience Rated Plans	Coefficient	Std Error	t-stat	p-value	CI Lower	CI Upper
Constant	4.667	0.021	221.402	~0	4.625	4.708
ACA_State	0.681	0.024	28.132	~0	0.636	0.729
Post_ACA	0.303	0.029	10.318	~0	0.245	0.360
ACA_x_Post	-0.020	0.035	-0.567	0.570	-0.088	0.048

Results of Regression (With Controls)

Experience Rated Plans	Coefficient	Std Error	t-stat	p-value	CI Lower	CI Upper
Constant	7.199	0.025	285.530	0	7.150	7.248
ACA_s	-0.335	0.025	-13.335	~0	-0.385	-0.286
$Post_t$	0.472	0.027	17.642	~0	0.420	0.524
ACA_s * $Post_t$	0.083	0.032	2.614	0.009	0.021	0.145
Inflation	4.978	0.779	6.389	0.00000000016	3.451	6.505
Inflation Annual Change	-3.850	0.643	-5.985	0.0000000021	-5.110	-2.589
S&P Yearly Growth	0.003	0.0004	7.261	~0	0.002	0.004
Federal Funds Rate Average Yield	-6.681	0.696	-9.593	~0	-8.045	-5.316

Results of Regression (With Controls)

Non-Experience Rated Plans	Coefficient	Std Error	t-stat	p-value	CI Lower	CI Upper
Constant	4.251	0.030	139.548	~0	4.192	4.311
ACA_s	0.754	0.028	26.579	~0	0.698	0.810
$Post_t$	0.592	0.035	16.972	~0	0.524	0.660
ACA_s * $Post_t$	-0.154	0.038	-4.011	0.00006	-0.229	-0.079
Inflation	6.979	0.930	7.503	~0	5.156	8.802
Inflation Annual Change	-1.813	0.746	-2.431	0.0152	-3.275	-0.351
S&P Yearly Growth	0.0001	0.0005	0.326	0.7445	-0.0009	0.001
Federal Funds Rate Average Yield	8.207	0.766	10.720	~0	6.707	9.708

Interpretation of Results (Regression Without Controls)

EXPERIENCE RATED PLANS:

ACAs (β1): -0.209368

- ACA States had 20.94% lower premiums per person than non ACA states before the implementation of the ACA.
- Significant at p-value < 0.05

ACA_s * $Post_t$ (β3): -0.092507

- After the ACA implementation premiums per person in ACA states decreased by an additional 9.25% compared to non-ACA states.
- The 95% confidence interval is [-0.149558, -0.035456] meaning that we can assume a negative effect.
- Significant at p-value < 0.05

NON-EXPERIENCE-RATED PLANS

ACAs (β1): 0.680978

- ACA states had 68.10% higher premiums per person than non-ACA states before the ACA was implemented.
- Significant at p-value < 0.05

ACA_s * $Post_t$ (β3): -0.019680

- Not statistically significant (not distinguishable from 0 at p-value < 0.05)
- 1.97% decrease in premiums per person for ACA states after the implementation of the ACA.

Overall: For Experience Rated Plans there is evidence that premiums per person significantly decreased and for Non-Experience-Rated Plans there seems to have been no effect from the implementation of the ACA.

Interpretation of Results (Regression with Controls)

EXPERIENCE RATED PLANS:

ACA_S (β1): -0.335350

- ACA states had 33.54% lower premiums per person than non-ACA states before the implementation of the ACA.
- Statistically significant and 95% confidence interval is [-0.384639, -0.286061] suggesting a significant negative effect.

ACA_S * $Post_t$ (β3): 0.082578

- After the ACA implementation premiums per person in ACA states increased by 8.26%.
- Statistically significant with the 95% confidence interval: [0.020664, 0.144492] suggesting a significant positive effect.

NON-EXPERIENCE-RATED PLANS

ACA_S (β1): 0.754043

- ACA states had 75.40% higher premiums per person than non-ACA states before the ACA implementation.
- The 95% confidence interval is [0.698440, 0.809646] suggesting a very significant positive effect.

ACA_S * $Post_t$ (β3): -0.153902

- After the ACA implementation premiums per person in ACA states decreased by 15.39% compared to non ACA states.
- Statistically significant with the 95% confidence interval being [-0.229108, -0.078695].

Overall: For experience rated plans the ACA increased premiums per person while for Non-Experience-Rated Plans it decreased them, both results are significant.